

KERKI, Turkmenistan

WMO: 389110

Lat: 37.833N

Lon: 65.200E

Elev: 787

StdP: 14.28

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.2	23.9	10.4	9.6	29.2	15.2	12.2	37.0	27.6	59.1	23.7	56.3	5.7	300	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.5	105.5	72.0	103.1	70.9	100.8	70.1	76.5	100.2	74.6	97.9	72.8	95.7	9.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.5	107.6	93.0	66.3	99.9	90.9	64.3	92.8	87.6	40.6	100.5	38.8	98.2	37.0	95.8	89.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.6	18.8	16.3	13.8	110.3	9.0	1.8	7.4	111.5	2.1	112.6	-3.0	113.6	-9.5	114.8		
(5)			12.9	79.3	8.7	4.6	6.6	82.6	1.5	85.3	-3.4	87.8	-9.8	91.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.6	41.2	44.8	55.7	66.7	77.5	84.9	87.7	84.3	74.6	63.1	51.2	42.7		
	DBStd	18.20	9.46	10.13	9.07	8.59	6.11	4.55	3.94	4.78	5.78	7.74	9.13	8.49		
	HDD50	892	294	195	48	3	0	0	0	0	0	6	92	254		
	HDD65	2949	738	565	311	83	4	0	0	0	4	127	425	692		
	CDD50	6234	20	50	224	506	851	1047	1168	1064	739	412	126	27		
	CDD65	2816	0	0	22	135	390	597	703	599	293	68	9	0		
	CDH74	38012	0	4	212	1387	4754	8696	10498	8188	3377	805	90	1		
	CDH80	22114	0	0	46	526	2469	5381	6738	5032	1655	246	21	0		
(14) Wind	WSAvg	7.1	7.6	8.4	8.4	7.5	7.1	7.3	7.7	6.9	5.7	5.8	6.2	6.8		
(15) Precipitation	PrecAvg	7.1	1.2	1.0	1.9	1.1	0.5	0.0	0.0	0.0	0.0	0.1	0.5	1.0		
	PrecMax	13.1	3.8	2.6	4.4	3.2	1.5	0.6	0.0	0.0	0.2	1.0	2.7	2.4		
	PrecMin	3.8	0.0	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.2	0.8	0.6	1.0	0.8	0.4	0.1	0.0	0.0	0.0	0.2	0.7	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.2	73.2	85.8	95.2	101.8	107.1	109.5	107.1	99.8	90.1	83.1	69.8		
		MCWB	49.3	51.8	61.9	67.7	68.9	71.4	75.1	73.3	68.0	64.0	60.4	50.3		
	2%	DB	62.5	67.9	80.1	90.0	98.1	104.1	106.4	104.0	96.0	86.1	75.5	64.7		
		MCWB	48.1	51.5	58.5	65.1	68.7	70.4	73.4	70.8	66.8	61.6	56.3	49.1		
	5%	DB	58.5	63.7	75.4	86.0	95.0	101.5	103.7	101.3	92.7	82.5	70.6	60.4		
		MCWB	46.8	50.2	56.8	63.9	68.0	70.2	72.0	69.8	65.9	60.3	54.2	47.9		
	10%	DB	54.8	59.9	71.0	82.3	92.1	99.0	101.3	98.6	89.3	78.4	66.3	55.9		
		MCWB	45.4	48.7	55.3	62.4	66.4	69.2	70.7	68.5	63.9	58.5	52.5	46.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	56.5	65.5	71.9	75.7	78.0	79.5	77.1	71.7	66.0	61.5	54.6		
		MCDB	62.1	65.2	82.0	91.7	96.0	100.4	103.2	101.4	93.5	86.9	81.0	63.0		
	2%	WB	50.4	54.0	61.4	68.6	73.0	75.9	77.0	74.5	69.2	63.1	58.1	51.1		
		MCDB	58.9	63.4	74.0	85.1	92.3	98.2	102.2	98.3	91.7	82.6	72.1	61.2		
	5%	WB	48.1	52.0	58.8	66.3	70.7	74.0	74.8	72.2	67.1	61.2	55.6	48.9		
		MCDB	56.5	61.4	71.0	81.2	89.6	96.5	99.7	95.8	89.6	80.0	67.6	58.5		
	10%	WB	45.8	49.7	56.5	64.2	68.6	72.1	72.6	70.2	65.2	59.3	53.4	46.7		
		MCDB	53.7	58.7	68.0	78.0	86.8	94.3	97.0	94.4	86.6	76.6	64.6	55.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	17.4	19.3	21.3	23.9	26.1	26.5	27.9	27.9	25.1	20.1	16.4		
		MCDBR	21.8	23.3	25.2	26.5	28.2	29.5	29.2	30.0	30.7	29.1	26.2	22.9		
		MCWBR	11.2	11.3	10.1	9.8	10.0	10.3	10.8	11.1	12.1	12.5	12.1	11.7		
		MCDBR	19.2	20.5	22.5	24.3	25.4	26.5	28.0	28.7	28.8	26.8	23.4	20.3		
	5% WB	MCWBR	11.3	11.6	11.3	10.9	11.5	12.1	12.5	12.5	13.0	12.7	12.3	11.9		
(40) Clear-Sky Solar Irradiance	taub		0.394	0.437	0.499	0.517	0.513	0.512	0.491	0.460	0.439	0.476	0.450	0.399		
	taud		2.085	1.983	1.860	1.857	1.886	1.877	1.923	1.994	2.012	1.897	1.951	2.059		
	Ebn at Noon		237	242	240	245	249	248	252	258	254	225	213	224		
	Edh at Noon		38	48	60	63	62	63	60	55	50	51	43	36		
(44) All-Sky Solar Radiation	RadAvg		684	957	1343	1805	2264	2572	2512	2300	1907	1345	849	634		
	RadStd		75	102	110	158	85	51	41	41	42	78	80	74		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.45	+2.39	+2.95	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	+0.58	-0.65	-0.90	N/A	N/A	N/A	N/A

Nomenclature: See separate page